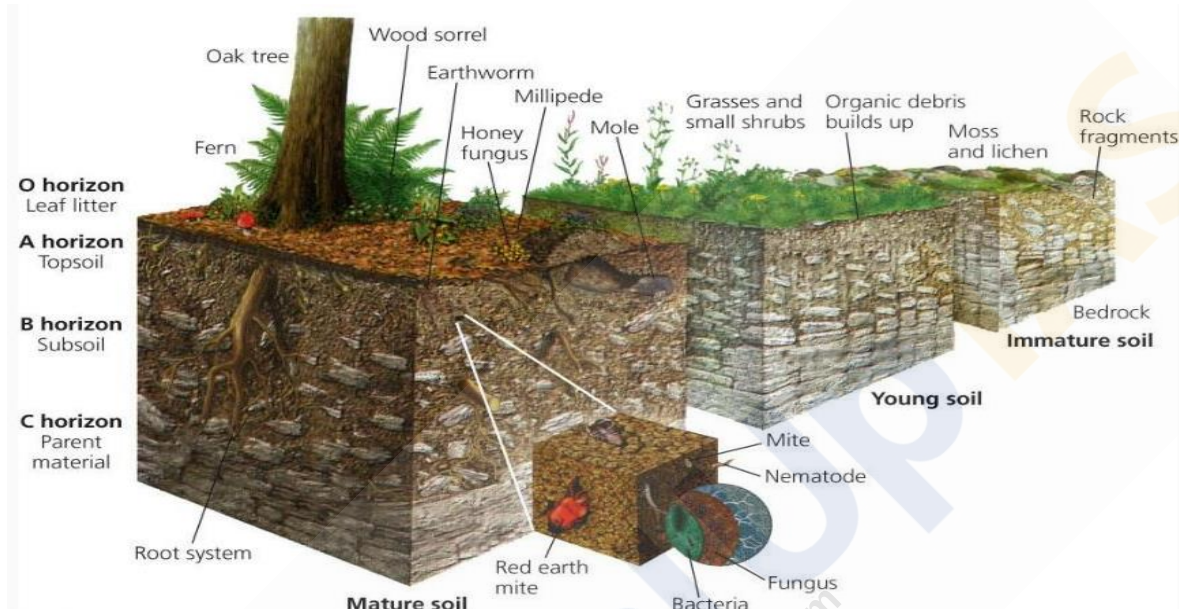


WORLD CLIMATE

- Soil: Composition of organic & inorganic particles.
- Formation: Physical factors (climatic conditions)
 - Geological factors (Parent rock)
 - Biological factors (Plants/animals/micro-organisms)
 - Mechanical factors (thawing)



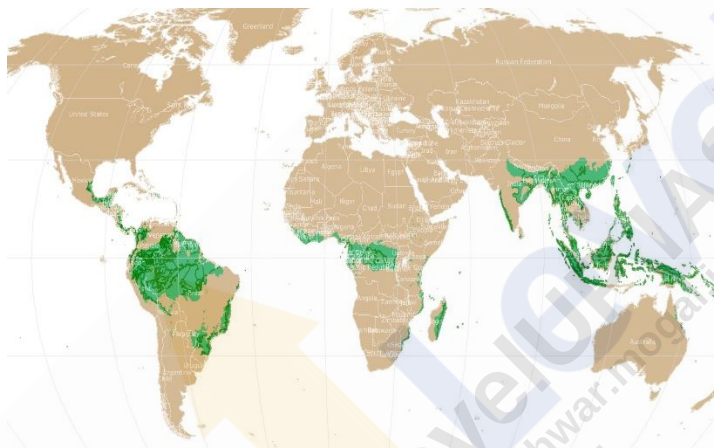
Major types of soils:

- (i) Alluvial soils:
 - Constitutes 46% of the area.
 - Silt deposits found in Indo-gangetic-Brahamaputra region
 - Immature & weak profile
 - Porus in nature – loamy structure
 - Less nitrogen
 - Adequate potash & phosphorus acid
 - Suitable for agriculture.
- (ii) Black soils:
 - Volcanic in origin
 - Constitutes 16% of total area
 - Present even in Tamil Nadu
 - Highly argillaceous (clayey)
 - High moisture retention
 - Self-fertilising & self-ploughing in nature
 - Low Phosphates, nitrogen & humus.

- (iii) Red soils: → Derived from metamorphic & crystalline
 → Constitutes 10% of area.
 → Mainly acidic in nature
 → Rich in potassium & poor in nitrogen & humus
 → Mostly loamy, retain water like black soils
- (iv) Laterite soils: → End products of weathering
 → Found in high temperature & high rainfall regions
 → Rainfall promotes leaching of silica
 → Rich in bauxite / ferric oxides
 → High phosphate content in form of iron phosphates
 → Known for plantation crops
 → Provide valuable building material.

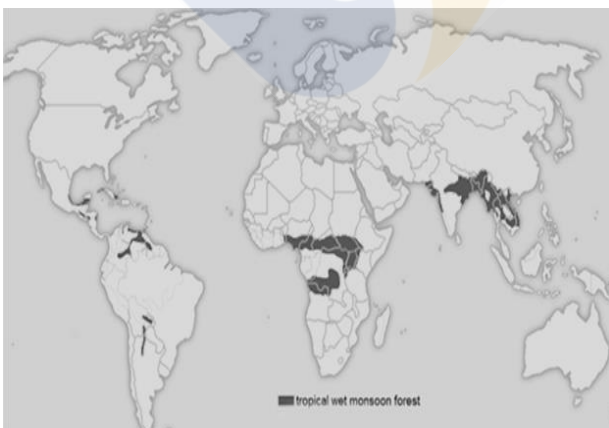
TYPES OF WORLD CLIMATE

Tropical evergreen climate (or) Rainforest Climate:



- Less annual range of temperatures
- No seasonal demarcation
- Heavy precipitation, thunderstorms with afternoon rainfall.
- Highly dense
- Also called as **Selvas**.
- Presence of epiphytic structures.
- Less undergrowth
- Shifting cultivation is highly practised
- Acidic soils (leaching process)
- Mostly found in basins of Amazon, Congo, Indo-Malasia

Monsoon climate



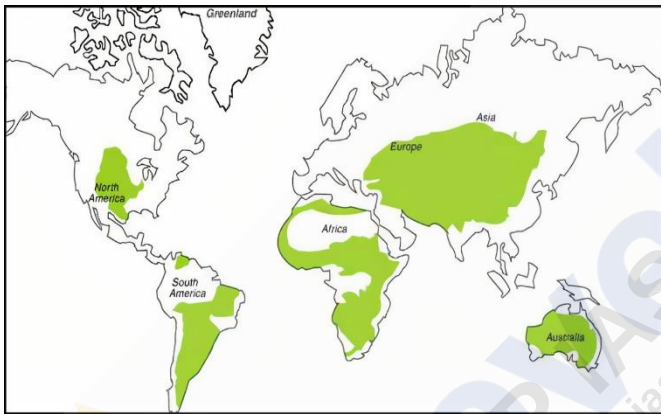
- Seasonal reversal of winds
- Clear demarcation of seasons
- Found between 5°-30° N & S
- Differential heating & concentrated heavy rainfall
- known for tropical deciduous forests
- High population density
- Known for plantation crops

Savanna Climate



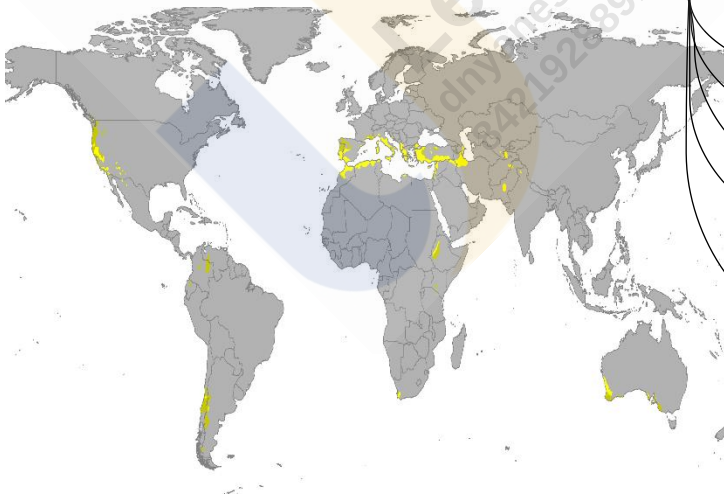
- Less annual rainfall
- No distinct rainy season
- Rains in summer
- Also called as **Sudan type**
- Known with extreme diurnal ranges
- Tall grass & short trees
- Known for Big game hunting
- Not suitable for Ranching activity
- Called as Llanos & Campos in S.America

Steepe Climate (or) Temperate grasslands



- Lies in the interior of the continent
- Extreme conditions in Northern hemisphere
- Chinook is present
- Shorter grass & trees less
- Known for ranching activity
- Called as **the Granaries of the world**

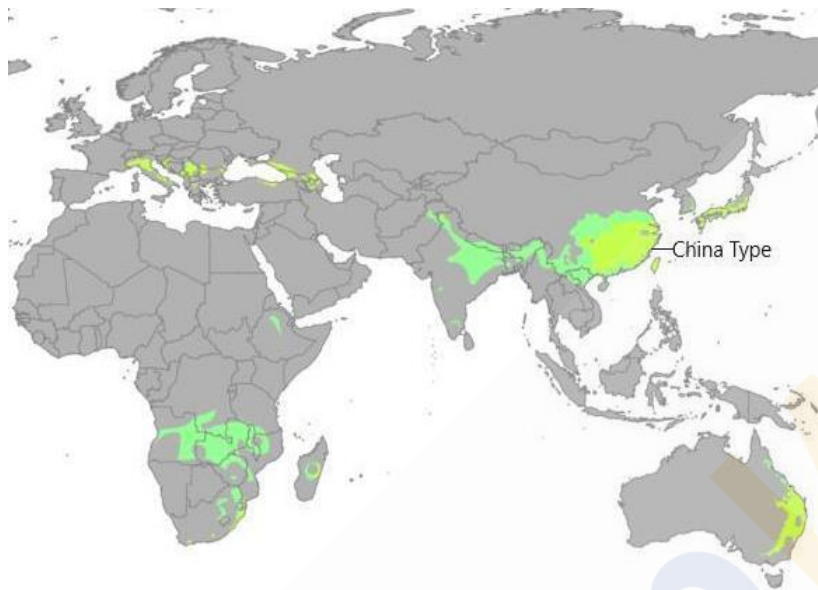
Mediterranean climate



- Also called as Western Margin climate
- Wet winters & Dry summers (shifting of pressure belts)
- Trees with small broad leaves
- Absence of shade
- Contain evergreen coniferous trees
- Known as **Orchids of the world & Wine yards of the world.**
- Sirocco & Mistral are present.

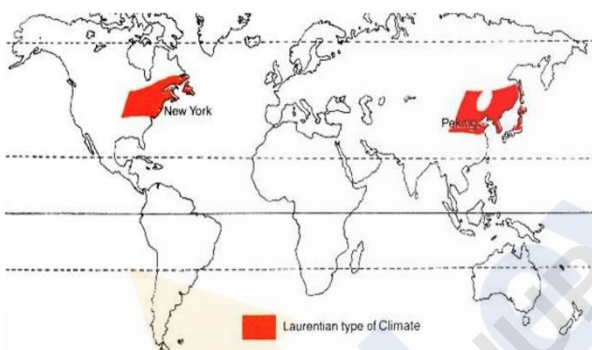
China type climate (or) temperature monsoon climate

- Warm moist summers & cool dry winters
- Gulf type of climate & natal type of climate are similar



Laurentian Climate

- Absent in Southern hemisphere
- Feature of both maritime & continental climate
- Intermediate between British type & Taiga type
- Cold dry winters & wet warm winters
- Cool temperate forests
- Known for large scale fishing

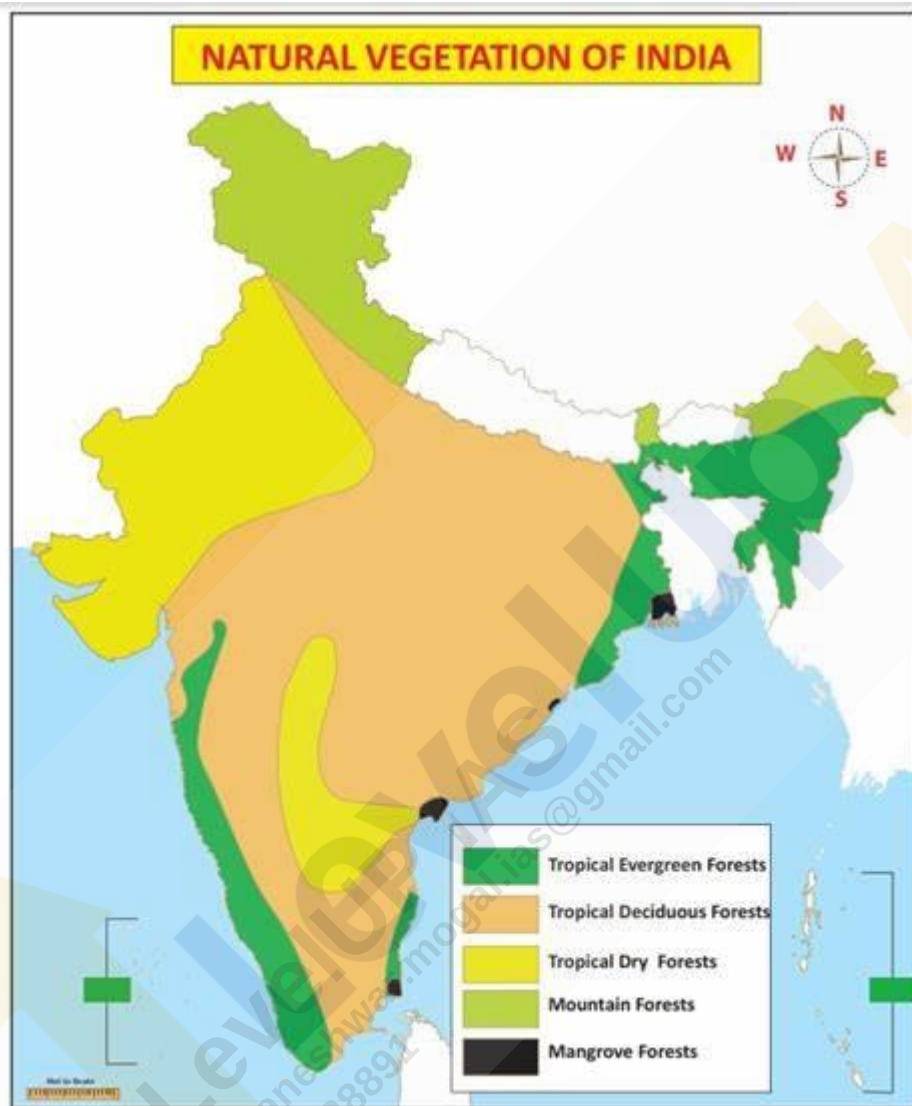
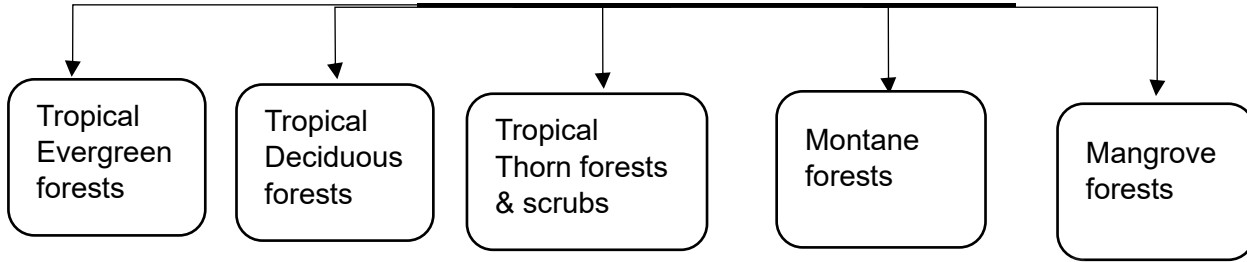


Tundra Climate

- Low mean annual temperature
- Snow & sleet form of precipitation
- Found in North Arctic circle & south of Antarctic circle
- No trees
- Known for rhododendrons, mosses, lichens



NATURAL VEGETATION OF INDIA



Tropical Evergreen forests:

- > 200 cm rainfall with short dry season.
- 18-30° C
- Height of trees is upto 60 mtrs
- Epiphytic structures
- Minimal undergrowth
- Found at Western ghats, Lakshadweep, Andaman & Nicobar & Tamil Nadu coast
- Flora: Ebony, mahogany, rosewood, rubber and cinchona.
- Fauna: Elephant, monkey, lemur and deer. One horned rhinoceros in Assam & West Bengal

Tropical Deciduous Forest:

- 70-200 cm rainfall
- Also called as Monsoon Forest
- Moist Deciduous – 100-200cm rainfall
- Dry deciduous – 70-100 cm rainfall
- Found in eastern part of the country, N-E states, foothills of Himalayas, Jharkhand. Chattisgarh, slopes of Western ghats.
- Shed leaves early summers & springs (Moist)
- Flora: Teak, Bamboos, sal, Khair, Kusum, Arjun, Mulberry
- Fauna: Lion, tiger, pig, deer and elephant

Thron forests & Scrubs:

- <70cm rainfall
- 25°-30° C temperature
- Scattered and long rooted trees
- Succulent stems
- Thick leaves
- Found in N-W part of country (Rajasthan, Gujarat, MP, Chhattisgarh, UP, Haryana)
- Flora: Acacias, palms, euphorbias and cacti
- Fauna: rats, mice, rabbits, fox, wolf, tiger, lion, wild ass, horses and camels.

Montane Forests:

- Wet temperate: 1000-2000 mtrs height, broad-leaf tress (Oaks, chestnuts)
- Temperate: 1500-3000 mtrs height, coniferous tress (pine, deodar, cedar, spruce)
- Flora: Mosses, Lichens
- Fauna: Kashmir stag, spotted deer, wild sheep, jack rabbit, Tibetan antelope, snow leopard

Mangrove Forests:

- 100-300 cm Rainfall
- 26°-35° C temperature
- Mangroves are salt tolerant plants
- Adapted to harsh coastal conditions
- Found at Deltas of ganga, Mahanadi, Krishna, Godavari, Kaveri.
- Flora: Palm, coconut, keora, agar
- Fauna: Royal Bengal tiger, turtles, crocodiles, gharials, snakes.

